

PAYSTREAM — COMPANY BRIEF

Payment Processing FinTech | London, UK | March 2026

COMPANY SNAPSHOT

PayStream Financial Ltd

- Size: 89 employees
- Sector: FinTech — Merchant Payments Platform
- HQ: Shoreditch, London
- Founded: 2019
- Focus: API-first payment processing for UK SME and mid-market merchants
- Regulatory: FCA-authorised Payment Institution (PI000012345)

Our Business: PayStream operates a payment gateway that handles merchant transactions (card, bank transfer, buy-now-pay-later). We process £1.2bn in annual transaction volume for 4,200+ active merchants. Merchants pay transaction fees; we keep 1.5% of each successful payment.

Your Division: Analytics & Operations Team (6 people)

- Your role: Junior Data Analyst (first hire in this team's expansion)
 - Manager: Sarah Ahmed (Senior Ops Manager)
 - Peers: 1 Data Engineer, 3 Operations Specialists, 1 QA Analyst
-

THE URGENT PROBLEM — THIS WEEK

Current Situation

As of **Monday 24 March 2026**, PayStream's transaction failure rate hit **12.1%** — a significant jump from our historical baseline of 5.8%.

This means:

- Out of every 100 transactions, 12 fail to complete
- Merchants lose revenue (they're charged our fee but receive no payment)
- Customers abandon shopping carts (friction = churn)
- Reputation risk: merchants are calling our support team

The Cost

Weekly Impact: ~£49,000 in lost merchant confidence + failed transactions **Cumulative (4 weeks):** £196,000 in Q1 targets at risk

Sarah Ahmed has told senior leadership: *"We'll fix this in 2 weeks or lose 8-10 key merchant partners."*

The Deadline

Hard stop: Friday 4 April 2026 — end of Q1
This is when Board reviews Q1 KPIs and performance bonuses are calculated.

If failure rate isn't below 4.5% by then:

- FCA regulatory concern (transaction accuracy reporting)
- Merchant churn projection: 12%
- Revenue impact: £180K+ in lost fees over 3 months
- Your team loses headcount approval for H2 expansion

YOUR MISSION IN 6 PHASES (2 WEEKS)

Week	Phase	Your Deliverable	Deadline
Week 1 (Mar 24-28)	Phase 2: Discovery	SQL analysis of failed transactions	Wed 26 Mar
	Phase 3: Root Cause	Written findings report with root cause %	Fri 28 Mar
	Phase 4: Recommendation	Suggested remediation options	Fri 28 Mar
Week 2 (Mar 31-Apr 4)	Phase 5: Stakeholder Communication	Executive summary & data briefing	Mon 31 Mar
	Phase 6: Business Decision	Final recommendation to engineering team	Thu 4 Apr

PAYSTREAM'S 4 PERSONAS

1. Sarah Ahmed — Your Manager

Role: Senior Operations Manager

Pressure: She owns the SLA. If we miss the 4.5% target, she explains to the Board why payments

failed.

What she needs: Clear analysis showing exactly which transactions are failing and why. Specific numbers she can cite.

Personality: Direct, data-driven, no patience for vague analysis. She'll ask: *"What percentage of failures come from X cause?"*

Sarah's Inbox Messages:

- Monday 24 Mar: "We have a crisis. Our failure rate jumped from 5.8% to 12.1% overnight. Need to know by EOD Wednesday which merchant types and payment methods are affected. Check the queue data."
 - Wednesday 26 Mar (after your Phase 2): "Good. API_Direct is the culprit. Now I need the root cause. Is it a rule misconfiguration? Timeout? Get me percentages by Thursday EOD."
 - Friday 28 Mar (after your Phase 3): "Right. 43% of API failures are from our payment validation rule. That's fixable. Send me 3 options for remediation—cost and timeline for each."
 - Monday 31 Mar: "Engineering is asking merchant questions. I'm briefing them Tuesday. Need your analysis deck and talking points."
 - Wednesday 2 Apr: "We're recommending Option 2 to the Board. Make sure your data is bulletproof. They'll ask for audit trail."
-

2. James Norton — Data Engineer (Your Peer)

Role: Data Engineer, owns the data pipeline

Pressure: If the data itself is corrupted or delayed, the whole analysis fails. He's paranoid about data quality.

What he needs: Confirmation that the data I provided is clean and the failure counts are accurate. Specific row IDs or date ranges to validate.

Personality: Cautious, asks follow-up questions, wants you to show your work.

James's Inbox Message:

- Thursday 27 Mar: "I see you're querying the failed_transactions table. Before you report those numbers, confirm: are you excluding the null week (week 9 data seems incomplete). And validate: row count for API_Direct should be ~8,400 across 12 weeks. Does your query match?"
-

3. Priya Patel — Head of Compliance (Stakeholder)

Role: FCA Compliance Officer

Pressure: Every payment platform must report transaction accuracy monthly to FCA. If our failure rate isn't explained and remediated, FCA will audit us.

What she needs: Clear documentation of the root cause and the plan to fix it. She needs this for

regulatory filing.

Personality: Formal, regulatory-minded, asks about processes and controls.

Priya's Inbox Message:

- Friday 28 Mar: "Sarah copied me on the remediation plan. Before we go to the Board, I need written confirmation: (1) what % of failures are from the validation rule, (2) what's the fix, (3) how we'll prevent it next time. This has to go in the FCA report."
-

4. Marcus Lee — Chief Product Officer (Stakeholder)

Role: Sets product direction, owns merchant experience

Pressure: Merchants are his customers. High failure rate = merchant churn = revenue.

What he needs: Confidence that we understand the problem and have a real fix. He'll decide whether to notify merchants or keep quiet.

Personality: Pragmatic, wants simple numbers. Cuts through complexity.

Marcus's Inbox Message:

- Monday 31 Mar: "I have merchants asking why payments are failing. Need the 30-second version: (1) what went wrong, (2) when will it be fixed, (3) will it happen again. Your data briefing needs to answer these."
-

THE DATA YOU INHERIT

Files Available to You (from Day 1)

1. payment_transactions_queue.csv (12 weeks of transaction data)

- What: Every payment attempt logged since 12 January 2026
- When: Updated daily, latest data from 23 March
- Columns: week, merchant_id, merchant_type, payment_method, transaction_count, failed_transactions

2. merchant_directory.csv (merchant master list)

- What: List of all 4,200+ active merchants with attributes
- Columns: merchant_id, merchant_name, merchant_type, sector, signup_date, monthly_volume

3. payment_methods_glossary.md (internal reference)

- What: Definitions of payment method codes and processing rules
- Contains: Payment_method codes, FCA classification, processing rules (including the validation rule flagged by Priya)

4. `paystream_sprint_plan.md` (what engineering is doing)

- What: Engineering sprint roadmap
- Shows: What changes were deployed in Week 8 that might have caused the spike

PAYSTREAM'S KEY METRICS

Metric	Current	Target	Timeline
Failed Transaction Rate	12.1%	4.5%	By 4 April (11 days)
Merchant Satisfaction	68%	85%+	By 31 May
FCA Compliance Score	At Risk	Clean	By 15 April filing
Monthly Processed Volume	£102M	£110M+	By 30 April
API Uptime	99.2%	99.8%+	By 30 May

GLOSSARY OF PAYSTREAM TERMS

Term	Definition	Used In Data
API_Direct	Merchants integrate directly into PayStream API (high volume, lower margin)	payment_method column
Merchant_Portal	Merchants use PayStream's hosted checkout (easier, lower risk)	payment_method column
Bank_Transfer	ACH/Faster Payments (lower failure rate, slower settlement)	payment_method column
BVR	Billing Validation Rule — checks merchant account is funded before processing	implicated in failures
SME	Small-Medium Enterprise (100–5,000 employees)	merchant_type column
Mid-Market	5,000–50,000 employees	merchant_type column
Enterprise	50,000+ employees	merchant_type column
Failed_pct	% of that week's transactions that failed	calculated metric

Term	Definition	Used In Data
Root Cause %	What percentage of ALL failures originated from a single cause	key finding you'll calculate

REGULATORY CONTEXT (FCA)

PayStream's Obligations:

- Monthly reporting to FCA on transaction accuracy and failure rates
- Immediate notification if any single cause accounts for >40% of failures
- Remediation plan required within 10 days of discovery

Priya's Red Line: "If our failure rate is 12.1% and 43% of those are from a single preventable cause, FCA will require immediate intervention. You need to prove the cause, prove the fix works, and prove it won't happen again."

SPRINT TIMELINE

Date	Event	What Happens
Mon 24 Mar	Failure rate spike detected	Sarah briefs team. Crisis mode.
Wed 26 Mar	Your Phase 2 analysis due	Sarah needs to know: which payment method, which merchant type?
Fri 28 Mar	Your Phase 3 root cause due	Sarah briefs engineering with your root cause %. James validates data.
Fri 28 Mar	Your Phase 4 options due	Engineering estimates fix timeline and cost.
Mon 31 Mar	Your Phase 5 stakeholder briefing	Marcus (Product) and Priya (Compliance) attend. You present.
Tue 1 Apr	Engineering implements fix	Code deployed to production.
Wed 2-3 Apr	Monitor & measure	Failure rate monitored in real-time.
Thu 4 Apr	Board review + FCA filing deadline	Sarah reports Q1 KPIs. If <4.5%, success. If not, explanation required.

WHAT SUCCESS LOOKS LIKE

- ✓ **Phase 2:** Correctly identify that API_Direct payments from SME merchants are the primary failure source (not Merchant_Portal or Bank_Transfer)
 - ✓ **Phase 3:** Calculate that 43.2% of failures are from the Billing Validation Rule (BVR) misconfiguration deployed in Week 8
 - ✓ **Phase 4:** Propose 3 options: (1) Revert BVR rule (£0, fixes in 1 hour), (2) Adjust rule thresholds (£12K engineering, fixes in 8 hours), (3) Add manual override queue (£45K, fixes in 3 days)
 - ✓ **Phase 5:** Brief stakeholders with data, visuals, and answers to their specific concerns
 - ✓ **Phase 6:** Recommend Option 1 (immediate revert) with justification using your calculated figures
-

WHAT FAILURE LOOKS LIKE

- ✗ Report vague findings like "payments are failing" without naming payment_method or merchant_type
 - ✗ Calculate a root cause percentage but get the denominator wrong (% of failed transactions vs. % of all transactions)
 - ✗ Miss the Week 8 anomaly (deployment trigger) and blame "random variation"
 - ✗ Present numbers without audit trail (James will ask: "Which table did you query? What's your row count?")
 - ✗ Propose solutions without understanding the cost/benefit tradeoff that Sarah and Marcus need to make
-

YOUR FIRST TASK (Starts Tomorrow, 25 March)

Phase 2 Deliverable Due: Wednesday 26 March, 5 PM

What to do:

1. Open payment_transactions_queue.csv
2. Write SQL queries to answer:
 - Which payment_method has the highest failure rate?
 - Which merchant_type is most affected?
 - What's the trend over the 12 weeks? (Did it spike suddenly or gradual?)
3. Submit your SQL queries + findings

Success metric: Sarah gets 3 specific numbers:

- "API_Direct has a 37% failure rate vs. 2% for other methods"
- "SME merchants are 8x more affected than Mid-Market"
- "Failure rate jumped from 6% to 18% between Week 8 and Week 9"

Then Phase 3 begins: *Why?*

END OF BRIEFING

You now understand:

- The business crisis and 11-day deadline
- The 4 people who will judge your work and what they need
- The data available to you
- The regulatory context
- What success and failure look like

Next: Your Phase 1 onboarding module will include the complete files package (CSVs, glossary, sprint plan).

You start tomorrow. Let's go.